Dead neurons resurrect by an entry–exit law: the smoothed-ReLU limit of training dynamics at an activation boundary

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Abstract

For shallow softplus networks under gradient flow we show that, in the sharp limit $\beta \to \infty$ of the activation stiffness, the death and resurrection of a neuron is governed by a hybrid system *with memory*: the dead neuron's parameters freeze, and reactivation occurs when the signed time integral of the network residual returns to zero — the classical entry–exit function of slow–fast dynamics — with rate $O(\beta^{-1})$. The switching face carries an identically zero field on the dead side, a degeneracy outside generic piecewise-smooth exit theory and precisely of the exponentially flat type of Kristiansen; the coupling of the forcing to the flow itself (the residual is generated by the very trajectory it gates) has, to our knowledge, no analogue in the entry–exit literature. The law is independent of the regularization within monotone sigmoid classes, including the erf class whose regularized activation is GELU. All constants are verified numerically; companion pre-registered experiments probe the stochastic gate under SGLD.

1 Setup and hypotheses

Data $(x_i, y_i)_{i=1}^n$ in general position, shallow network $f_\theta(x) = \sum_k g_\beta(w_k x + b_k)$ with $g_\beta(z) = \beta^{-1}\log(1 + e^{\beta z})$ (softplus; see §5 for general sigmoids), quadratic loss $L = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$, gradient flow $\dot\theta = -\nabla L$. Fix a neuron A and a datum j; write $z = \theta_A \cdot (x_j, 1)$ for its margin, $c = \|(x_j, 1)\|^2/n$, and $R(t) = y_j - f_{-A}(x_j; \theta_B(t))$ for the residual *excluding* A. The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb}, $A^{hyb}(t) = \int_{t_{hit}}^t c R^{hyb}$.

Hypotheses. (H1) $R^{hyb}(t_{hit}) < 0$ (transversal death; this also gives the transversal arrival, Lemma 3). **(H2)** A^{hyb} has a first zero at $t_{exit} < T$ with $R^{hyb}(t_{exit}) > 0$ and $A^{hyb} < 0$ on (t_{hit}, t_{exit}). **(H3)** the other margins of A stay $\le -\delta_1 < 0$ on the episode.

2 Main result

Definition 1 (hybrid limit)**.** *θ^∞ equals: the ReLU flow on $[0, t_{hit}]$; on $[t_{hit}, t_{exit}]$, $\theta_A \equiv \theta_A(t_{hit})$ (frozen) and θ_B the B-only flow; on $[t_{exit}, T]$, the ReLU flow restarted at $(\theta_A(t_{hit}), \theta_B^{hyb}(t_{exit}))$.*

Theorem 1 (T2′-min)**.** *Under (H1)–(H3), for every $\delta > 0$,*

$$\sup_{t \in [0,T]\setminus(B_\delta(t_{hit}) \cup B_\delta(t_{exit}))} \|\theta^\beta(t) - \theta^\infty(t)\| = O_\delta(\beta^{-1}), \qquad t^\beta_{release} = t_{exit} + O(\beta^{-1}).$$

The limit is that of the regularized flow: discrete ReLU gradient descent freezes the dead neuron forever (the limits do not commute).

3 Proof outline

Six lemmas; each has an executable verification script in the repository.

Lemma 1 (normal form)**.** *In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$: $dw/d\tau = cR(t)\, w^2/(1+w)\,(1+E_\beta)$, $|E_\beta| \le \log(1+w)/(\beta|R|)$ — the quadratic normal form of De Maesschalck–Schecter with $\varepsilon = \beta^{-1}$. With $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact quadrature $d\Phi/dt = -c\beta R(t)(1+E_\beta)$.*

Lemma 2 (B-subsystem and feedback)**.** *On any episode interval where $w \le w_{\max}$, the coupling of A into the B-field is $\le \beta^{-1}\log(1+w_{\max})$ and $\sup\|\theta_B^\beta - \theta_B^{hyb}\| \le C\beta^{-1}$ (Grönwall). In the bulk of sticking $u = O(\beta)$, so the coupling there is $O(\beta^{-2})$.*

Lemma 3 (entry corner)**.** *$\dot z^0(t_{hit}^-) = cR^{hyb}(t_{hit}) < 0$ by (H1); by comparison in the window $z \in [\beta^{-1}\log w_0, \delta_z]$, the crossing of $\{w = w_0\}$ occurs at $t_{hit} + O(\beta^{-1})$ with state $O(\beta^{-1})$-close to $\theta^\infty(t_{hit})$.*

Lemma 4 (sticking and release, prescribed R)**.** *For prescribed R satisfying (H1)–(H2): $\Phi(u(t)) = \Phi(u_0) - c\beta A(t)(1+o(1))$, hence $u(t) = c\beta|A(t)|(1+o(1))$ (depth $z \sim -\beta^{-1}\log\beta$), and u returns to $O(1)$ at $t_{exit} + O(\beta^{-1})$ — no $\log\beta$ factor. Sub/super solutions from the monotonicity of the flow in R give explicit constants; perturbations δ of R shift t_{exit} by $\le c\,\delta\,(t_{exit} - t_{hit})/|A'(t_{exit})| + O(\delta^2)$.*

Lemma 5 (coupled bootstrap)**.** *The set of times up to which "θ_B is $2M\beta^{-1}$-close to the hybrid and the margin stays in the layer" is open and closed in $[t_{cross}, t_{exit} - \epsilon]$: under the assumption, coupling is $O(\beta^{-1})$ (corners) / $O(\beta^{-2})$ (bulk), so Lemma 2 returns the better constant $M\beta^{-1}$ and Lemma 4 with $\delta = O(\beta^{-1})$ keeps the layer. Hence the closed-loop law $t_{release}^\beta = t_{exit} + O(\beta^{-1})$.*

Lemma 6 (exit corner; scalar reduction)**.** *Post-release the field is transversal outward ($cR^{hyb}(t_{exit}) > 0$, (H2)); gluing as in Lemma 3. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$ and the face-j field is rank one along $(x_j, 1)$, so the tangential components of θ_A freeze to exponential accuracy and the episode reduces to the scalar problem.*

4 Numerical verification

Instance $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$; $t_{hit} = 0.1114$, $t_{exit} = 1.1325$, $\delta_1 = 0.508$. Selected checks (scripts `passo0`–`passo6`): entry $t_{cross} = 0.11141 + 7.13/\beta$ (fit residual 0.2% over $\beta \in [3\cdot 10^2, 10^4]$); depth ratio $u_{\max}/(c\beta|A|_{\max}) \to 0.98$; bulk feedback $\times\beta^2 \approx 43$ constant over 1.5 decades; closed-loop $\beta\,(t_{release}^\beta - t_{exit}) \to -14.4$ (a definite first-order coefficient, sharper than the theorem); tangential drift of θ_A at machine zero.

5 Regularization independence; discussion

The release criterion depends only on $\int R\,dt = 0$, not on the sigmoid: for the erf class ($\sigma = \Phi_{\text{gauss}}(\beta z)$, whose activation is regularized GELU) the coupled system releases at the same t_{exit} with $O(\beta^{-1})$ gap (verified). Open: the cone apex (corank ≥ 2), where weight decay produces

genuine sliding toward the most singular stratum; the SGLD gate (pre-registered experiments show the deterministic schedule is falsified at first order by persistent residual fluctuations — the stochastic entry–exit is the object of the next cycle); and the non-commutation with discrete-time ReLU training.

Provenance

Every lemma has an executable symbolic/numeric verification committed alongside; experiments were pre-registered and adversarially reviewed before execution; two earlier designs and one earlier conjecture were rejected by that process and are archived in the repository.

References

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